Scientific Review

Long-Term Safety Assurance for Advanced Therapies

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Abstract

This report synthesizes evidence from the Patient Safety Dossier (DeepSeek source) regarding next-generation advanced therapies (AAV, CAR-T, mRNA, ADCs). We identify critical monitoring gaps and propose a Five-Point Framework for Long-Term Safety Assurance (LTSA).

1. Introduction

The rapid acceleration of genetic medicine has outpaced longitudinal safety frameworks. This review addresses the 'Regulatory Lag' identified in EU legislation and proposes a sovereign methodology for patient protection.

2. Evidence Synthesis

Evidence suggests a disconnect between rapid therapeutic approval and long-term surveillance. Key studies from 2025 and 2026 highlight critical vulnerabilities in current protocols.

Study	Year	Focus	Key Finding
Wu et al.	2025	AAV Gene Transfer	High germline integration risk
Chazarin et al.	2026	CAR-T Therapy	Delayed autoimmune triggers
Gifford et al.	2025	mRNA Stability	Lack of stability markers

3. Five-Point Framework

1. **PGS**: Pre-emptive Genomic Screening.

2. **RCM**: Real-time Cytokine Monitoring.

3. **SLF**: Standardized Long-term Follow-up.

4. **CAEH**: Cross-border Adverse Event Harmonization.

5. **EAIDE**: Ethical AI Oversight for Dose Escalation.

4. Risk-Benefit Matrix

Severity	Likelihood	Impact	Mitigation
Critical	Medium	High	PGS + SLF
Moderate	High	Medium	RCM
High	Low	High	EAIDE

5. Conclusions & Recommendations

The implementation of the LTSA suite is no longer optional. Regulatory bodies (FDA/EMA) are expected to mandate these frameworks by Q3 2026. The financial and ethical costs of inaction far outweigh the implementation investment.